Poslední úkol – řešení

① Jsou dány body $A[1,2,2]$, $B[3,1,2]$, $C[5,1,1]$. Určete

a) rovnici roviny $\alpha = (A,B,C)$,
b) vektor $\vec{x}$ délky 12 kolmý k α,
c) vzdálenost $D[4,-1,2]$ od α.

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a) $\overrightarrow{AB} = B - A = (2,-1,0)$, $\quad \overrightarrow{AC} = C - A = (4,-1,-1)$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 0 \\ 4 & -1 & -1 \end{vmatrix} = (1,2,2)$$

$$\left.\begin{array}{l} \alpha\colon\ x + 2y + 2z + d = 0 \\ A \in \alpha\colon\ 1 + 2\cdot 2 + 2\cdot 2 + d = 0 \ \Rightarrow\ d = -9 \end{array}\right\} \quad \boxed{\alpha\colon\ x + 2y + 2z - 9 = 0}$$

b) $\vec{x} = t \cdot \vec{n}_\alpha = t(1,2,2) = (t,2t,2t)$

$$12 = \|\vec{x}\| = \sqrt{t^2 + (2t)^2 + (2t)^2} = \sqrt{9t^2} = 3t \ \Rightarrow\ t = 4, \qquad \boxed{\vec{x} = (4,8,8)}$$

c)

1) bodem D kolmice k k α: $\quad k\colon\ x = 4 + t,\ \ y = -1 + 2t,\ \ z = 2 + 2t$
2) $k \cap \alpha$:

$$\begin{aligned} 4 + t + 2(-1 + 2t) + 2(2 + 2t) - 9 &= 0 \\ 4 + t - 2 + 4t + 4 + 4t - 9 &= 0 \\ 9t &= 3 \\ t &= \tfrac{1}{3} \end{aligned} \qquad P\left[\tfrac{13}{3}, -\tfrac{1}{3}, \tfrac{8}{3}\right]$$

3) vzdálenost D od $\alpha = \left\|\overrightarrow{PD}\right\|$:

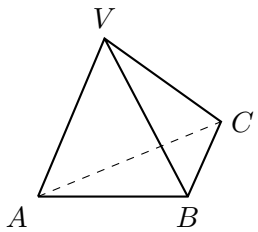
$$\overrightarrow{PD} = D - P = [4,-1,2] - \left[\tfrac{13}{3}, -\tfrac{1}{3}, \tfrac{8}{3}\right] = \left(-\tfrac{1}{3}, -\tfrac{2}{3}, -\tfrac{2}{3}\right)$$

$$\left\|\overrightarrow{PD}\right\| = \sqrt{\left(-\tfrac{1}{3}\right)^2 + \left(-\tfrac{2}{3}\right)^2 + \left(-\tfrac{2}{3}\right)^2} = \sqrt{\tfrac{9}{9}} = 1$$

vzdálenost D od α je $\boxed{1}$

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② Objem a výška čtyřstěnu $ABCV$: $\quad A[3,5,3]$, $B[-2,11,-5]$, $C[1,-1,4]$, $V[0,6,4]$



$$\overrightarrow{AB} = B - A = (-5,6,-8)$$
$$\overrightarrow{AC} = C - A = (-2,-6,1)$$
$$\overrightarrow{AV} = V - A = (-3,1,1)$$

$$V = \frac{1}{6}\left|\left[\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AV}\right]\right| = \frac{189}{6} = \boxed{31{,}5} \quad \ldots \text{ objem}$$

$$\left[\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AV}\right] = \begin{vmatrix} -5 & 6 & -8 \\ -2 & -6 & 1 \\ -3 & 1 & 1 \end{vmatrix} = -5\cdot(-6)\cdot 1 + (-2)\cdot 1\cdot(-8) + (-3)\cdot 6\cdot 1 -$$
$$- (-8)\cdot(-6)\cdot(-3) - 1\cdot 1\cdot(-5) - 1\cdot 6\cdot(-2) =$$
$$= 30 + 16 - 18 + 144 + 5 + 12 = 189$$

$$V = \frac{1}{3} S_p \cdot v \;\Rightarrow\; v = \frac{3V}{S_p} \overset{(*)}{=}$$

$$S_p = S_{\triangle ABC} = \frac{1}{2}\left\|\overrightarrow{AB} \times \overrightarrow{AC}\right\| = \frac{1}{2}\sqrt{(-42)^2 + 21^2 + 42^2} = \frac{1}{2}\sqrt{3969} = \frac{1}{2} \cdot 63$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -5 & 6 & -8 \\ -2 & -6 & 1 \end{vmatrix} = (-42, 21, 42)$$

$$(*) = \frac{3 \cdot \frac{189}{6}}{\frac{1}{2} \cdot 63} = \boxed{3} \;\ldots\; \text{výška}$$

③ Vzdálenost rovnoběžných rovin α a β: $\quad \alpha\colon x - 3z + 2 = 0, \quad \beta\colon 2x - 6z - 7 = 0$

$\vec{n}_\alpha = \vec{n}_\beta = (1, 0, -3)$

1) $A \in \alpha$ libovolný … $A\left[0, 0, \frac{2}{3}\right]$
2) $k \perp \alpha$, $A \in k$ … $k = A + t\,\vec{n}_\alpha$: $\quad x = t, \quad y = 0, \quad z = \frac{2}{3} - 3t$
3) $k \cap \beta = B$:

$$2t - 6\left(\tfrac{2}{3} - 3t\right) - 7 = 0$$
$$2t - 4 + 18t - 7 = 0$$
$$20t = 11 \;\Rightarrow\; t = \tfrac{11}{20}$$

$B\left[\frac{11}{20}, 0, \frac{2}{3} - \frac{33}{20}\right]$

4) vzdálenost α a $\beta = \left\|\overrightarrow{AB}\right\|$:

$$\overrightarrow{AB} = B - A = \left[\tfrac{11}{20}, 0, \tfrac{2}{3} - \tfrac{33}{20}\right] - \left[0, 0, \tfrac{2}{3}\right] = \left(\tfrac{11}{20}, 0, -\tfrac{33}{20}\right)$$
$$\left\|\overrightarrow{AB}\right\| = \sqrt{\left(\tfrac{11}{20}\right)^2 + 0^2 + \left(-\tfrac{33}{20}\right)^2} = \sqrt{\tfrac{1210}{400}} = \sqrt{\tfrac{121}{40}} = \boxed{\frac{11}{2\sqrt{10}}}$$